

Partial-Map Intervention Laboratory (PMIL)

Laboratory Design, Mission Charter, and Rigorous Framework

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Document Control

Executive Summary

The Partial-Map Intervention Laboratory (PMIL) is a simulation-design and research framework for studying decisions made under incomplete knowledge of an internal system layout. Its central problem is not simply flow. It is the choice among observing, probing, intervening, rerouting, escalating, or stopping when the architecture, current state, and risks are only partly known.

PMIL was designed as an independent laboratory within the broader Control Flow Matrix research family. It retains the foundational principle “recognition of layout before intervention,” while separating several related but non-identical problem types: access under uncertainty, return-path or control failure, reachable-state recognition, architecture comparison, and temporal lineage propagation.

The Version 1 program uses synthetic systems and tabletop scenarios. It is not a medical decision-support product, does not prescribe vascular-access actions, and does not collect or infer personal biological, genetic, or neural information. Its purpose is to develop durable simulation architecture, transparent uncertainty handling, and cross-domain design methods.

Mission

To develop durable, uncertainty-aware simulation environments that help agents recognize layout, represent incomplete knowledge, compare safe action alternatives, and learn from outcomes before escalating intervention in complex systems.

PMIL will improve simulation design by making architecture, state, observation quality, policy limits, intervention costs, and history first-class objects rather than hidden assumptions.

Purpose and Objectives

- Create a reusable representation for partially observable systems without assuming that all domains obey the same physics.
- Model the distinction between a route that does not exist and a route that has not yet been detected.
- Test the value of observation before intervention when probing, failure, delay, or damage carry costs.
- Separate physical feasibility from permitted action by modeling policy, consent, risk, operator capability, and stopping rules.
- Provide a safe synthetic environment for comparative design across biological-style, engineered, computational, and fictional systems.
- Create an auditable basis for later modules involving transport, state reachability, architecture comparison, and temporal propagation.
- Preserve different viewpoints and alternative models instead of requiring premature convergence on one theory or outcome.

Foundational Principle

Recognition of layout before intervention.

Operationally, PMIL treats a controller as an agent working with incomplete observations. The agent estimates a system state, chooses whether further observation is worth its cost, selects an allowed action, observes the result, and updates its model. The

laboratory contrasts this loop with blind trial-and-error.

Observe → Infer → Choose → Act → Measure → Update
Not recommended: Guess → Act → Fail → Retry

Problem Families

Problem family	Representative inspiration	Core laboratory question	Primary model
Access under uncertainty	Difficult blood draw analogy	Does an acceptable route exist, and is the current evidence sufficient to act safely?	Partial observability; belief-state estimation; costly sensing
Return-path/control failure	Varicose-vein analogy	What occurs when a pathway remains present but directionality, control, properties, or history degrade?	Dynamic transport; failure history; gates and feedback
Reachable-state recognition	Epoch Blades	Can a current configuration reach a target configuration through allowed, acceptable transitions?	State-space reachability; trajectory planning
Architecture comparison	Roots, leaf venation, pipes, networks	Which similarities survive abstraction, and which do not transfer?	Typed graph comparison; invariant testing
Temporal propagation	Generational/lineage thought experiment	How do information and transformations propagate through time?	Separate temporal-lineage graph; later module

These families are related, but PMIL does not reduce them to one primitive. A node-edge diagram can represent a route, but it cannot by itself represent an inaccessible interface, a policy prohibition, uncertainty about a route, accumulated damage, or the reachability of a target state.

Scope and Non-Goals

- In scope: synthetic networks, tabletop exercises, reusable data schemas, uncertainty representation, intervention policies, comparative simulation, and fictional state-transition systems.
- In scope: domain adapters that define the correct local meaning of quantities such as driving potential, resistance, capacity, state, and control.
- Out of scope for Version 1: real patient-specific vascular guidance, clinical recommendations, diagnosis, real neural-signature analysis, predictive claims about an individual's learning limits, or real genetic identification and comparison.
- Out of scope for Version 1: a universal equation claiming that blood, information, electricity, genetics, learning, and social systems have identical governing dynamics.
- Out of scope for Version 1: merging all possible domains into one full-scale atlas before the shared representation has been tested.

Core Definitions

Term	PMIL operational definition
Matrix	The structured embedding context in which entities exist, connections are realized, state changes occur, and one or more media may move, transform, or be constrained.
Architecture	Nodes, edges, compartments, interfaces, geometry, and connectivity available to the system.
Medium	What moves or transforms: fluid, charge, packet, resource, signal, or abstract state.
State	The current values and conditions of entities, edges, controls, and environment.
Driving potential	A domain-specific difference or input that causes movement or state change; it is not assumed to be physical pressure in every domain.
Observation	Measurements available to the agent, including confidence, provenance, timing, and limitations.
Uncertainty	What is unknown, estimated, ambiguous, unmeasured, or unreliable; unknown is never treated as absent.
Control	Mechanisms that regulate routing, access, transformation, or response.
Policy	Rules defining which physically feasible actions are permitted, prohibited, escalated, or stopped.

Term	PMIL operational definition
History	Prior states, loads, interventions, damage, adaptations, and observed outcomes.
Reachability	Whether a target state can be reached from a current state under allowed transformations, constraints, and acceptable cost.

Conceptual Framework

PMIL uses a meta-model rather than a universal physical law. Each domain supplies its own governing equations, rules, or transition logic. At the comparison layer, the laboratory requires a common set of declared variables:

$$Behavior = f(Architecture, State, Properties, Driving Conditions, Control, Environment, History, Observation, Uncertainty, Policy)$$

The expression is a checklist for model completeness. It does not state that all systems share the same mechanism or mathematics. Domain adapters decide whether a field applies and, if it does, how it is measured and interpreted.

Layer	Question answered
0. Identity	What specific system, domain, scale, and epistemic status are being modeled?
1. Architecture	What exists, and what connects to what?
2. Properties	What capacities, impedances, materials, gates, sensors, and redundancies characterize the structures?
3. State	What is happening now?
4. Driving conditions	What inputs, gradients, demands, or forces cause change?
5. Observation	What can be measured, with what confidence and limits?
6. Control	What regulates behavior or routing?
7. Policy	Which otherwise possible actions are permitted, prohibited, or subject to escalation?
8. Intervention	What action is selected, at what expected cost and risk?
9. Response	What happened after action?
10. Learning	How should the model, belief state, policy, or future action change?

Laboratory Architecture

PMIL is designed as a modular program rather than a monolithic simulator. A shared Architecture Representation Layer supplies durable objects and metadata; specialized laboratories implement domain-specific models without forcing false equivalence.

- Architecture Representation Layer: common identities, schemas, provenance, epistemic labels, uncertainty fields, and audit history.
- Partial-Map Intervention Laboratory: observation planning, belief-state estimation, action selection, stopping rules, escalation, and outcome learning.
- Transport and Return-Path Laboratory: routing, capacity, resistance, gates, directionality, degradation, feedback, and failure propagation.
- State Reachability Laboratory: current state, target state, allowable transformations, constraints, costs, reachability, and trajectory evaluation. The Epoch Blade model belongs here as a fictional system.
- Architecture Comparator: typed graph comparison, architecture fingerprints, transferability analysis, and explicit analogy-failure reporting.
- Temporal Lineage Laboratory: a later, separated module for inheritance, transformation, recombination, population history, and time-based propagation using synthetic data first.

Data and Model Schema v0.1

Every model must declare its domain, epistemic status, scope, variables, and limitations. Fields that do not apply may be explicitly omitted rather than faked.

Schema block	Required content
System	Identifier, name, domain, scale, purpose, time model, epistemic status
Matrix	Embedding context, geometry when relevant, boundaries, environmental conditions, architecture
Architecture	Nodes, edges, compartments, interfaces, directionality, connectivity
Medium	Type, conservation behavior where relevant, transformation rules
State	Node, edge, controller, system, and environmental state at a defined time
Driving conditions	Typed domain-specific potential, demand, gradient, or rule-defined input
Properties	Capacity, impedance/resistance, material properties, redundancy, thresholds
Control	Sensors, controllers, gates, feedback loops, control laws
Observation	Measurements, confidence, provenance, measurement method, unknowns, observability
Policy	Permitted and prohibited actions, consent rules, attempt limits, stopping and escalation rules
History	Prior states, interventions, accumulated load, damage, repair, adaptation
Intervention	Action, target, expected information gain, expected cost, risk, constraints
Outcome	Success and failure metrics, state change, uncertainty change, lessons learned

Two fields are mandatory for PMIL experiments: observation cost and intervention cost. A pathway can exist while remaining expensive to observe, unsafe to probe, inaccessible at present, or disallowed by policy.

Epistemic and Ethics Framework

Label	Meaning
Established	Supported by relevant real-world evidence and used within stated limits.
Domain model	A deliberately simplified representation for a simulation.
Hypothesis	A testable proposal that requires evaluation.
Analogy	A comparison used for reasoning; it is not evidence of identical mechanisms.
Fictional rule	A setting-specific mechanism defined by narrative design.
Unknown	Insufficient evidence or data to classify, estimate, or use reliably.

- Start with synthetic, public, anonymized, or fully consented material; Version 1 requires no individual biological, genetic, or neural data.
- Never convert missing observation into a claim of absence.
- Use the minimum information needed for the stated simulation purpose.
- Distinguish simulation, education, hypothesis generation, and real-world deployment; moving between them requires separate validation and governance.
- Treat detailed individual representations as a custody and governance issue, not merely a technical asset.
- For fictional systems, specify ownership, consent, refusal, archival access, deletion or revocation, and misuse safeguards as part of worldbuilding.

Initial Experiment Program

The first cycle tests whether the schema can represent different kinds of uncertainty and intervention without flattening their differences.

Experiment	Synthetic setup	Decision question	Primary measures
A. Last Allowed Access Attempt	Venous-style branching graph with hidden routes, uncertain observations, probe costs, and a fixed escalation policy.	Observe, attempt, escalate, or stop?	Failed attempts; total cost; information gained; harm avoided; calibration of belief.
B. Degraded Return Path	Transport network with directional gates, capacity, resistance, history-dependent degradation, and alternate routes.	Repair, reroute, observe, or isolate?	Pooling/congestion; unmet demand; propagation; recovery; resilience.
C. Almost-Matching Welder	Fictional abstract state graph with archive states, allowed transformations, constraints, and cost.	Is an acceptable trajectory to the target form available?	Reachability; path length; transition cost; uncertainty; refusal conditions.
D. Leaf After Midrib Disruption	Synthetic branching/looped leaf-like network with a central edge weakened or removed.	How do redundancy and topology alter local isolation and redistribution?	Connectivity retained; delivery loss; cascade size; recovery time.

The first three experiments are sufficient for a Version 1 tabletop. The fourth supplies an early comparator case. All experiments should report not only where a similarity appears, but also which mechanisms do not transfer.

Evaluation Framework

- Model fidelity: Does the model represent its declared domain without hidden or false equivalences?
- Uncertainty discipline: Does it distinguish absent, unobserved, estimated, and confirmed conditions?
- Policy realism: Are feasible actions separated from permitted actions?
- Intervention efficiency: Does additional observation reduce expected harm or cost when it should?
- Calibration: Are confidence estimates consistent with eventual outcomes?
- Comparative rigor: Are cross-domain similarities labeled structural, dynamic, analogical, or non-comparable?
- Reproducibility: Can scenarios, inputs, policies, and outputs be re-run and audited?
- Governance: Are data provenance, consent boundaries, ethical status, and model limitations visible?

Governance and Operating Model

PMIL should operate as an open-to-dissent design environment. It is intended to support different simulation missions, different hypotheses, and competing interpretations rather than forcing all collaborators into one conclusion.

- Every model receives a version number, preparer, date, scope note, epistemic label, and change log.
- Every scenario declares assumptions, omitted variables, intended interpretation, prohibited interpretations, and limitations.
- Alternative models may coexist when they use the same schema but different domain adapters or control rules.
- No model becomes canonical merely because it is visually compelling or cross-domain; it must identify its evidence, assumptions, and boundaries.
- Fictional and real-world-adjacent modules remain separately labeled and may exchange only explicitly defined abstractions.

Funding Pathways

Funding should be staged so that early work remains low-risk, synthetic-first, and focused on reusable simulation infrastructure. The pathways below are categories to investigate; eligibility, solicitations, and requirements change over time and should be verified before application.

Pathway	Fit with PMIL	Early deliverables that support pursuit
Internal or personal prototype funding	Best for schema design, tabletop exercises, documentation, and a small proof-of-concept.	Versioned charter, schema v0.1, synthetic scenario set, prototype backlog.

Pathway	Fit with PMIL	Early deliverables that support pursuit
University or community-college faculty collaboration	Useful for systems engineering, computer science, simulation, biology, education, or ethics mentorship.	Research question, scoped MVP, faculty-aligned concept note, student project plan.
NSF-style basic research programs	Potential fit for computational modeling, network science, cyber-physical systems, trustworthy AI, simulation, and interdisciplinary methods.	Clear novelty claim, evaluation plan, reproducible synthetic testbed, data-management and broader-impacts plan.
NIH/AHRQ-style health-systems or training research	Only for later validated educational, workflow, or simulation research; not for unvalidated clinical decision support.	Synthetic training simulator, clinician/educator partners, safety case, IRB strategy if human research is proposed.
DOE, national-laboratory, or infrastructure resilience programs	Potential fit for network resilience, control, digital twins, systems modeling, and failure propagation.	Engineered-network module, quantified resilience metrics, reproducible perturbation studies.
DARPA/ARPA-style high-risk research pathways	Potential long-term fit only after a strong technical core demonstrates a novel approach to partial observability, adaptive control, or simulation.	Demonstrated prototype, technical differentiation, risk-reduction milestones, responsible-use framing.
State and regional innovation programs	Potential support for workforce training, education technology, infrastructure simulation, or applied research partnerships.	Local partner statement, workforce or community benefit, implementation plan.
Philanthropic foundations and responsible-technology grants	Potential fit for ethical AI, transparent simulation, public-interest technology, education, and interdisciplinary research.	Public-interest mission, governance plan, accessibility and dissemination strategy.
Industry research partnerships	Potential fit for digital twins, simulation platforms, networks, logistics, visualization, or educational tools.	Non-exclusive research scope, IP terms, synthetic benchmark, partner-specific use case without compromising independence.

Funding safeguard: early proposals should emphasize synthetic simulation, formal uncertainty, transparent governance, and educational or infrastructure-research value. They should not promise clinical prediction, personal neuroprofiling, or genetic comparison capabilities.

Phased Roadmap

Phase	Duration target	Primary output	Exit criteria
0. Charter and vocabulary	2–4 weeks	This design charter; controlled glossary; scope boundaries	Terms are stable enough to support a schema.
1. Schema v0.1	2–6 weeks	Machine-readable and narrative representation schema	Three distinct systems can be described without forced fields.
2. Tabletop scenarios	2–6 weeks	Synthetic experiments A–C with decision logs	Scenarios distinguish observation, intervention, policy, and uncertainty.
3. Minimal prototype	1–3 months	Small interactive or scripted simulator using synthetic graphs	Reproducible experiment runs and explicit outcome metrics.
4. Comparator pilot	1–3 months	Architecture-fingerprint and transferability report	Model labels structural/dynamic/an alogical/non-comparable similarity.
5. External review and partners	Ongoing	Technical, ethics, and domain review	Assumptions and boundaries survive independent critique.
6. Expansion decision	After validation	Go/no-go decision for transport, education, fiction, or lineage modules	MVP evidence justifies added scope.

No phase automatically authorizes the next. Advancement requires a documented review of model limits, ethics, reproducibility, and whether the prior phase has produced useful evidence rather than only more terminology.

Risks and Mitigations

Risk	Mitigation
Scope expansion	Use designable slices, phased gates, and a strict Version 1 domain set.

Risk	Mitigation
False equivalence across domains	Require domain adapters, epistemic labels, and analogy-failure reports.
Overconfidence from incomplete maps	Treat uncertainty as a first-class layer; require confidence and provenance.
Sensitive-data drift	Synthetic-first policy; no individual biological, genetic, or neural data in Version 1.
Clinical misinterpretation	Maintain non-clinical educational/simulation framing and prohibit treatment recommendations.
Fiction/reality confusion	Separate fictional rules from evidence-backed or modeled claims.
Unfundable breadth	Pursue modular funding tied to specific deliverables and validated milestones.
Opaque AI outputs	Require versioning, assumptions, audit logs, scenario replay, and explicit uncertainty.

Deliverables

- PMIL Charter and Laboratory Design v1.0.
- Controlled Vocabulary and Definitions v0.1.
- Architecture Representation Schema v0.1 in YAML/JSON and narrative form.
- Synthetic Scenario Pack v0.1: access-under-uncertainty, degraded return path, Epoch Blade reachability, and leaf disruption.
- Experiment Log Template with assumptions, policy, uncertainty, actions, outcomes, and lessons learned.
- Comparator Reporting Template for structural, dynamic, analogical, and non-comparable findings.
- Ethics, Data, and Fiction Boundary Memorandum.
- Prototype Product Requirements Document after tabletop validation.

Conclusion

PMIL is designed to make a difficult but general class of problems tractable: an agent must act in a system whose internal layout is variable, only partly visible, and potentially costly to probe. The laboratory does not claim that all systems are the same. It provides a disciplined way to compare how different systems handle partial observability, state, control, policy, history, and intervention.

Its first success criterion is modest and rigorous: demonstrate that a single typed schema can represent an uncertain access problem, a degraded return-path problem, and a fictional reachable-state problem while preserving the distinctions that make each problem meaningful. If it can do that, PMIL will have earned expansion into a broader control-flow and architecture research program.